

## Pre-class preparation

Please read the following, where the textbook section comes from Degroot and Schervish's *Probability and Statistics* (fourth edition):

- Introduction to sampling distributions: pdf on website
- Textbook: Section 8.2

## Objectives

By the end of the day's class, students should be able to do the following:

- State the definition of the sampling distribution of a statistic, and find the sampling distribution for a variety of common statistics.
- Compute probabilities based on the sampling distribution of a statistic.
- Explain how to use properties of the sampling distribution to quantify how close an estimator is to the unknown parameter.
- Give the PDF and CDF of the  $\chi^2$  distribution, and explain its relationship to the Gamma distribution. and exponential distribution.
- Calculate the mean, variance, moments, and MGF of the  $\chi^2$  distribution using properties of the Gamma distribution.

## Reflection Questions

Please submit your answers to the following questions to the corresponding Canvas assignment by 9:00AM:

1. We learned that if  $X_1, \dots, X_n | \mu, \sigma^2 \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ , then the sampling distribution of  $\bar{X}_n | \mu, \sigma^2 \sim N(\mu, \sigma^2/n)$ . Suppose now that  $n = 100$  and  $\sigma^2 = 1$ . Explain how to use this information along with the 68-95-99.7 rules to calculate the probability that the sample mean is within 0.2 of the true mean  $\mu$ .
2. True or false? The sampling distribution for every statistic is approximately Normal.
3. Use known properties of the Gamma distribution to briefly show that if  $X \sim \chi^2$  with  $m$  degrees of freedom, then  $\mathbb{E}[X] = m$  and  $\text{Var}(X) = 2m$ .
4. (Optional) Is there anything from the pre-class preparation that you have questions about? What topics would you like some more clarification on?